# Sidharth Kannan

408-466-5904 | sid.kannan11@gmail.com | Google Scholar | GitHub | Personal Website

Machine learning researcher focusing on agents for science and engineering

## Technical Skills

**Machine Learning**: PyTorch, JAX, Slime, Harness Engineering, Flow Matching, Weights and Biases
**Scientific Computing**: NumPy, SciPy, distributed training (DDP), SLURM
**Chip Design**: FPGA, Vivado, SystemVerilog, SPICE, Agents for chip design

## Education

**University of California, Santa Barbara** — Santa Barbara, CA
*Bachelor of Science in Computer Science, Bachelor of Science in Physics* — *Sep. 2022 – March 2026*
GPA: 3.94, *High Honors*

## Work/Research Experience

**Research Engineer** — March 2026-Present
*ChipAgents* — *Santa Clara, CA*

- Built domain-specific datasets for semiconductor tapeout for post-training (SFT, RLVR) of GLM 5.2
- Shipped an agentic analog mixed signal design verification toolchain now in use at Broadcom, Micron, and other tier-1 semiconductor companies.
- Implemented model routing, programmatic tool calling, and caching optimizations that reduced token use by 40%.

**Computational Physics Fellow** — June 2025-March 2026
*Los Alamos National Laboratory* — *Los Alamos, NM*

- Developed codes for modeling relativistic electron scattering in dilute gases for nuclear weapons physics application
- Building flow-based models for super-resolving observables in particle-in-cell simulations of fusion reactors (jointly with Jeong Lab at UCSB)

**Undergraduate Research Assistant** — October 2022 – Present
*University of California, Santa Barbara* — *Santa Barbara, CA*

*Machine Learning Researcher* — *June 2024-May 2025*

- Built flow based generative models for representation learning of early-universe dark matter density fields
- Building flow-based models for super-resolving observables in particle-in-cell simulations of fusion reactors (jointly with Los Alamos National Laboratory)

*Physics Based Computation Researcher* — *October 2022-May 2024*

- Built first ever hardware implementation of a higher order Ising machine, achieving state of the art on XOR-Satisfiability problems
- Implemented graph clustering algorithms to optimize device layouts and enable efficient hardware resource sharing, leading to 20% increase in device capacity

**Research Intern** — June 2024 - December 2024
*LIGO Laboratory* — *Riverside, CA*

- Built graph neural network based surrogate models for accelerating simulations of LIGO interferometer
- Achieved 800x speedups over traditional numerical methods, with minimal loss of fidelity
- Compared frequency and spatial domain simulation for laser cavities, improving simulation of thermal aberrations

**Flight Software Intern** — June 2023-September 2023
*Astranis Space Technologies* — *San Francisco, CA*

- Built data analysis tooling for disk space allocation on Astranis satellites
- Wrote Global Navigation Satellite System (GNSS) receiver interface for satellite localization during orbit raise

**Firmware Engineering Intern** — June 2022-September 2022
*Tenafe, Inc.* — *Campbell, CA*

- Designed, tested, and documented a lightweight software framework for verification of TenaFe's SSD controller.
- Wrote core modules for simulating error correction, buffers, Flash memory, read, write, erase operations

## Selected Personal Projects

**Mini-GPT** May 2026 - Present

- Implemented GPT-2 style language model trained on TinyStories, including BPE tokenizer, RoPE, multi-head attention, and more advanced techniques like MoE.

**Fermiacc** May 2026 - Present

- Building AI agents for autonomous theory generation and validation for high-energy physics, in collaboration with UCSB physicists.

## Publications

- **Kannan, S.**, Qiu, T., Cuesta-Lazaro, C., Jeong, H. (2025). Λ-CFM: Scale-Aware Representation Learning for Cosmology with Flow Matching. *Machine Learning: Science and Technology 2026.*, In Press.
- **Kannan, S.**, Goodarzi, P., Papalexakis, E., Richardson, J. (2025). Graph Neural Networks for Interferometer Simulation. *Neural Information Processing Systems 2025, AI4Science Workshop.*
- Nikhar, S.*, **Kannan, S***., Aadit, N. A.*, Chowdhury, S., Camsari, K. Y. (2024). All-to-all reconfigurability with sparse and higher-order Ising machines. *Nature Communications*, 15(1), 8977. **(* Equal contribution)**
- López-Paradís, G., Hair, I. M., **Kannan, S.**, Rabbat, R., Murray, P., Lopes, A., ... Balkind, J. (2024, June). The Case For Data Centre Hyperloops. In 2024 *ACM/IEEE 51st Annual International Symposium on Computer Architecture (ISCA)* (pp. 230-244). IEEE.

## Teaching Experience

**Undergraduate Learning Assistant** September 2023-Present

*University of California, Santa Barbara* *Santa Barbara, CA*

- Teaching assistant for "Intro to Scientific Computing" (1x), "Electrostatics" (1x), "Data Structures and Algorithms I" (2x), "Data Structures and Algorithms II" (2x), and "Discrete Mathematics" (1x)
- Held office hours and graded assignments

**Course Instructor** Santa Barbara, CA

*University of California, Santa Barbara*

- *CMPTG CS 5 - Statistical Physics and Neural Networks* - Taught a 10 week course on the relations between neural network theory and statistical physics, including generative models like energy based and diffusion models, and theoretical concepts like neural tangent kernels.
- *PHYS CS 5 - Non-equilibrium Statistical Mechanics*: Teaching 10 week course on non-equilibrium statistical mechanics. Topics covered include stochastic processes, Brownian motion, kinetic theory, and open quantum systems.

## Talks

- *Λ−CFM: Scale-Aware Representation Learning for Cosmology with Conditional Flow Matching*, Simons Foundation, Learning the Universe Collaboration Invited Talk, December 2025
- *Self-Consistent Relativistic Electron Scattering for X-Ray Diagnostics*, Kavli Institute for Theoretical Physics, UC Santa Barbara Undergraduate Research Symposium, September 2025
- *Statistical Mechanics of Machine Learning*, National Science Foundation Institute for Artificial Intelligence and Fundamental Interactions Reading Group, March 2025
- *Graph Neural Networks for Interferometer Simulations*, Kavli Institute for Theoretical Physics, UC Santa Barbara Undergraduate Research Symposium, September 2024

## Awards and Honors

- *NSF GRFP Honorable Mention*, honorable mention for the most prestigious graduate fellowship awarded by NSF
- *Regents' Scholar*, merit based scholarship awarded to top ∼2% of incoming undergraduates at UC Santa Barbara.
- *Semiconductor Research Corporation Research Scholar*, a 1-year undergraduate research fellowship awarded to ∼30 undergraduates nationwide. My fellowship was sponsored by IBM, and supported my work on probabilistic computers.